Visualisation tools: Pandas and Seaborn

Units 11–16 · syllabus 2024 · module marks 15, doubled to 30

HOW THIS PAPER WORKS

The syllabus gives this module 15 marks. This paper gives **twice** that (30 marks). Plots are drawn from the physics data named in the syllabus: capacitors, the convex-lens liquid-lens experiment, the sonometer, and the Iris data set.

No past paper of this exact course was found, so the questions use similar CUFYUGP papers. **PYQ** = adapted from a similar paper, named with its code, term and question number. **Instructor** = written from the syllabus, with no close match.

Similar papers used: MAT4CJ205 Python and SageMath (Apr 2026); STA3MN201 Statistical Inference using R (Nov 2025); STA2MN101 Probability Theory I (Apr 2025).

Time	2 hours
Maximum	30 marks

Section A: answer all ten. Section B: answer any two of three. Where a question asks for a plot, write the code and say what the plot shows.

Section A · Short answer

Answer all. Each question carries 2 marks. Total 20.

1. Give two reasons why a graph is more useful than a table of readings for finding a physical law. Readings taken at regular times (for example a capacitor voltage every second) are best shown as a time series. Write one line of seaborn code that draws `voltage` against `time` from a DataFrame `df`.

PYQ A26 Q4 · adapted · U11, U16

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2. Write the code that draws a bar chart of the mean voltage of three capacitors `C1, C2, C3` stored in a Series named `means`, with the title "Mean voltage" and axis labels.

PYQ STA3MN201 N25 A1 · adapted · U11

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3. A histogram of 200 readings of a pendulum period is drawn with `bins=10` and then with `bins=50`. Explain how the shape of the plot changes, and which choice is more honest about the spread of the data.

Instructor · U12

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4. What does a box plot show? Name the five numbers it summarises, and state what the length of the box represents.

Instructor · U12

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5. Write the code for a scatter plot of `length` against `period` with red circles of size 40 and the title "Pendulum". Name the keyword that sets the marker shape and the keyword that sets the marker colour.

PYQ A26 B14 · adapted · U12

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6. Explain how a correlation matrix plot is made from a DataFrame, and state what a value close to +1 in the matrix means for two columns `mass` and `length`.

PYQ STA2MN101 A25 B15 · adapted · U13

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7. What does `sns.set_palette('pastel')` change, and why would a physics student pick a colour palette that is safe for colour-blind readers?

Instructor · U14

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8. State the difference between a histogram and a kernel density plot. Which one is a smooth curve, and what does its area represent?

Instructor · U14, U15

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9. Write the code that draws a violin plot of `petal_length` for each `species` in the Iris data set, using a DataFrame `iris`.

Instructor · U15

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10. Explain what a hexbin plot shows and when it is preferred to a scatter plot. Then explain how a bubble chart differs from a scatter plot, and write one line of matplotlib code that draws it from arrays `x`, `y` and a size array `s`.

Instructor · U15, U16

Section B · Descriptive

Answer any two of the three. Each carries 5 marks. Total 10.

11. Liquid-lens experiment. Five readings of the refractive index of a convex lens filled with liquid are stored in a DataFrame with columns `trial` and `n`. Write code that draws a box plot of `n`, adds a title and axis labels, and saves the figure as `lens.png`. Then explain what an outlier in a box plot is, and how you would recognise one in this data.

Instructor · U12

12. Sonometer. The mass `M` and the length `L` are stored in the DataFrame `son`. Write code that (a) adds a column `L2 = L**2`, (b) draws a regression plot of `M` against `L2` with `sns.regplot`, and (c) prints the correlation matrix of `M` and `L2`. Explain what the slope of the line would mean physically.

Instructor · U13, U16

13. Iris data. Using the Iris set, write code that draws a pair grid of the four numerical columns, coloured by `species`, and a heatmap of the correlation matrix of the same four columns with the colour map `'coolwarm'`. Explain what a pair grid shows that a single histogram does not.

Instructor · U15, U16

COVERAGE

UNIT	TOPIC	QUESTIONS
11	Importance of visualisation, bar chart	1, 2
12	Histogram, frequency polygon, box plot, scatter plot, markers, labels, title	3, 4, 5, 11
13	Correlation matrix and correlation plot	6, 12
14	Seaborn features, colour palette, univariate distribution plot	7, 8
15	Seaborn histogram, density, bivariate plot, hexbin, violin, pair grid	8, 9, 10, 13
16	Bar plot for categories, regplot, heatmap, cmap, bubble chart, time series	1, 10, 12, 13

Total marks: 20 (Section A) + 10 (Section B) = 30. Every syllabus topic of units 11–16 is the main point of at least one question.